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Machine Learning for Effluent Forecasting at a Full-Scale Municipal Wastewater Treatment Plant: Seasonal Performance Deterioration and the Decisive Role of Prediction Horizon

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Abstract

Machine learning is widely proposed as a basis for smart wastewater treatment plant (WWTP) management, yet reported performance is rarely benchmarked against trivial alternatives. This study uses six years of monitoring from a municipal WWTP in Serbia serving 18,000 population equivalents, comprising 1017 fully measured days. Total nitrogen removal falls from 79.4% outside summer to 67.4% in June to August ($p = 4 \times 10^{-18}$), in five of six years, while influent load, hydraulic flow and phosphorus removal are unchanged, which points to oxygen limitation. Four model families are benchmarked against persistence and a ten-measurement rolling mean on one chronological hold-out period. Predictive skill depends decisively on forecast horizon: at one step ahead nothing beats both baselines, whereas at five steps nitrogen reaches R^2 of 0.516 against 0.388 and 0.320. Chemical oxygen demand (COD) never beats a rolling mean. A random forest identifies nitrogen exceedance of 15 mg/L with an area under the curve of 0.90. Explanation of that model shows it recognises a breach already under way rather than one beginning, which sets the operating conditions for its use. Evaluating on interpolated rather than measured days raises apparent accuracy from R^2 of 0.34 to 0.82.

Keywords: seasonal performance variation; nitrogen removal; effluent exceedance prediction; model benchmarking; baseline comparison; data imputation; reproducibility; wastewater treatment plant

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1. Introduction

Industrialisation and urbanisation have increased the pressure on water resources, highlighting the need for effective wastewater treatment and sustainable management of wastewater discharges [1–3]. Wastewater treatment plants (WWTPs) play a central role in protecting receiving water bodies [4], while their modernisation combines improved treatment performance with digital technologies that support process monitoring, automation, optimisation, and timely operational decision-making [5,6]. Ensuring proper treatment and responsible management of wastewater is of great importance for sustainable water resource management [1].

WWTP data can be complex, making it challenging to extract meaningful insights and make effective decisions [7]. Developing a suitable methodology for WWTP modeling and evaluating plant efficiency is crucial for optimising operations and achieving environmental goals. The application of advanced tools, including machine learning (ML) and deep learning (DL), in wastewater treatment is of increasing interest for performance prediction in WWTPs [8–10].

ML has been widely applied in wastewater treatment for water quality monitoring, predictive modeling, process optimisation, and WWTP management [11]. ML can process real-time data streams and predict trends that enable operators to make data-driven choices promptly [12]. ML models can simulate and predict time series data by capturing complex, often non-linear relationships between input and target variables from extensive observational data [13,14]. Researchers have recently applied ML techniques to forecast the quality of water resources, surface water and effluent water [15–19]. Wang et al. [20] conducted a comprehensive assessment of nine ML methodologies to simulate effluent quality parameters. Miao et al. [21] employed three ML algorithms, a Gated Recurrent Unit neural network, Support Vector Regression, and Long Short-Term Memory (LSTM) neural networks to replicate 60-day real-time outflow chemical oxygen demand (COD) concentration. Zaghloul and Achar [22] applied ML techniques to model full-scale WWTP with biological nutrient removal and to predict 15 process parameters. Overall, previous studies have focused on predicting WWTP performance using historical data collected over relatively short monitoring periods.

While numerous studies have applied ML to predict the performance of WWTPs, most have focused on improving prediction accuracy by selecting appropriate input variables and modeling techniques [6,7,11–14]. In comparison, the use of long-term operational data together with ML to support sustainable WWTP management has received much less attention [23,24]. Consequently, the potential of ML to support sustainable WWTP management by providing evidence for operational decisions and a more comprehensive evaluation of treatment plant performance remains insufficiently explored [23–25].

Despite the increasing number of studies applying advanced ML models, important methodological gaps remain in the existing literature. Most studies focus on a single prediction setting, providing limited insight into how model performance changes across different forecasting horizons, although such information is essential for practical operational decision-making. Also, only few studies based on full-scale, long-term WWTP datasets systematically examine seasonal variation in predictive performance, despite the temporal and operational variability characteristic of wastewater treatment processes [26].

With the aim of advancing current approaches to WWTP performance evaluation, this study combines long-term operational performance data with ML modeling and interprets the resulting findings from a sustainable WWTP management perspective.

The study makes three main contributions. First, it establishes a baseline-based evaluation framework that enables the added predictive value of more complex ML models to be assessed across different forecasting horizons and seasonal conditions. It further interprets the operational and modeling results in terms of their implications for sustainable WWTP management. Second, it demonstrates how long-term operational data can identify environmentally relevant performance patterns and support data-driven WWTP management. Third, it identifies the predictive configurations with operational value and the additional monitoring requirements needed for future Smart WWTP and Digital Twin applications.

2. Materials and Methods

2.1. Methodological Framework

The methodological framework was developed using a six-year operational dataset collected from a full-scale municipal wastewater treatment plant in Serbia. The framework consisted of four main stages: data preprocessing, statistical analysis, ML model development, and comparative evaluation of model performance. The resulting operational and modeling findings were subsequently interpreted in terms of their implications for sustainable WWTP management.

2.2. Plant Performance Description

The experimental data utilised in this research was obtained from a municipal WWTP designed for 18,000 population equivalents (PE), situated in the northern part of Serbia, in the province of Vojvodina. The wastewater treatment process in WWTP is designed to effectively treat a daily inflow of 9300 m^3 of raw wastewater, containing 3500 kg of biochemical oxygen demand (BOD_5). The aim is to ensure that the treated effluent meets the discharge standards set by legal regulations. The treatment methodology implemented at this WWTP is based on biological treatment employing activated sludge processes. The purification system comprises two distinct technological components: the water line and the sludge line. The water line employs a conventional purification process involving biologically active sludge. This method is utilised to treat the water stream, effectively removing contaminants and enhancing water quality. Concurrently, the sludge line implements an alternative process termed anaerobic digestion, conducted within the mesophilic temperature range. This specific range of temperatures optimises the efficiency of the anaerobic digestion process. As part of this process, the sludge undergoes mechanical dehydration, further concentrating the solid content and facilitating subsequent handling and disposal. By bifurcating the purification process into these two interconnected lines, the system can holistically address both water purification and sludge management. The water line targets the removal of pollutants and the improvement of water quality, while the sludge line adopts anaerobic digestion to break down organic matter and optimise sludge management. This dual-line configuration demonstrates a comprehensive approach to address both wastewater treatment and sludge management.

The treatment begins with the pumping station, where raw wastewater is collected from the sanitary sewer networks. Initial filtration is achieved through a mechanical grid, followed by the separation of sand and oil in a dedicated separator. The next step involves the primary clarifier, which facilitates the settling of solids and further purifies the wastewater. Aeration and recirculation of biologically active solids maintain diverse communities of microorganisms to degrade a wide range of organic compounds and nutrients. The aeration basin and secondary clarifier play significant roles in maintaining these microbial populations, with recirculation pumps enhancing the process.

To address the sludge generated during treatment, the primary thickener is employed to concentrate solids, which are then directed to the digester. This stage is essential for biogas production, serving to stabilise the sludge. The anaerobic digestion process occurs within a mesophilic temperature range, with a retention time of at least 20 days. The biogas produced in this phase is harnessed to sustain optimal temperature conditions for the mesophilic bacteria, ensuring the efficiency of the anaerobic digestion process. The sludge line is ended by a biogas tank and mechanical dehydration, where the sludge is further processed. The produced biogas is used to support the thermal requirements of the anaerobic digestion process. Ultimately, the treated wastewater is released into the Danube–Tisa–Danube canal.

2.3. Data Collection and Preparation

Records were obtained from the plant for the period January 2015 to July 2020 and comprise influent and effluent water temperature, pH, COD, BOD_5, total nitrogen (TN), total phosphorus (TP) and total suspended solids (TSS), together with wastewater flow (Q), hydraulic retention time (HRT), excess sludge and biogas production. Sampling was not performed on every calendar day: across the 2028-day span from 8 January 2015 to 27 July 2020, measurements were available for approximately half of the days. The median interval between consecutive measurements is 1 day, the mean is 2.0 days, and 72.7% of consecutive measurements fall within 2 days of one another.

2.4. Data Cleaning

Cleaning proceeded in six steps, and the number of surviving records after each is given below. The source file holds 1205 records. Removing duplicate daily entries leaves 1131. Removing records with any missing influent or effluent value leaves 1119. A flow plausibility filter leaves 1117, and removing records containing zero values leaves 1110. Effluent plausibility filters, COD below 125 mg/L and BOD_5 below 180 mg/L, leave 1075. Influent plausibility filters leave 1018, which reduce to 1017 distinct measurement days once repeat measurements on the same date are averaged. The influent filters were COD below 2000 mg/L, BOD_5 below 900 mg/L, TN below 200 mg/L, TP below 35 mg/L and TSS below 5000 mg/L. Flow records outside 1000 to 30,000 m^3/day were removed as physically implausible for a plant of this design capacity. Every threshold is a plant design or regulatory bound rather than a quantile of the observed data, so no record was excluded for being statistically unusual. The same cleaned frame is used for every analysis reported in this paper, and Figure 1 shows where it sits in the pipeline.

No imputation was performed. An earlier version of this analysis resampled the record to daily frequency and filled non-sampled days by time-weighted interpolation followed by a fifty-day trailing rolling mean. That procedure was abandoned because it introduces reconstructed values into both training and evaluation, and its effect is quantified in Section 3.7. In the analysis reported here every predictor and every target value is a measurement.

Influent parameters were standardised to zero mean and unit variance, with the scaler fitted on the training partition only and applied unchanged to the test partition, so that no information from the test period can reach the model through the scaling. Lagged effluent variables were retained in their original units so that reported errors are directly interpretable in mg/L. Mass loads were computed as concentration multiplied by daily flow. No logarithmic or power transformations were applied, because the models compared here are either scale invariant or fitted on standardised inputs, and because transforming the target would have made the reported errors harder to read against the regulatory limits expressed in mg/L.

For each observation, the ten preceding measurements of effluent and influent concentration for the target parameter were supplied as predictors, together with their arithmetic mean. Lags are defined over consecutive measurements rather than consecutive calendar days, which avoids introducing synthetic values where sampling was not performed. Given the observed sampling interval, a ten-measurement window corresponds to approximately three weeks of plant operation. Together with the seven same-day influent parameters, this yields 28 predictors. Observations lacking a complete lag history were discarded, leaving 1007 modeling records.

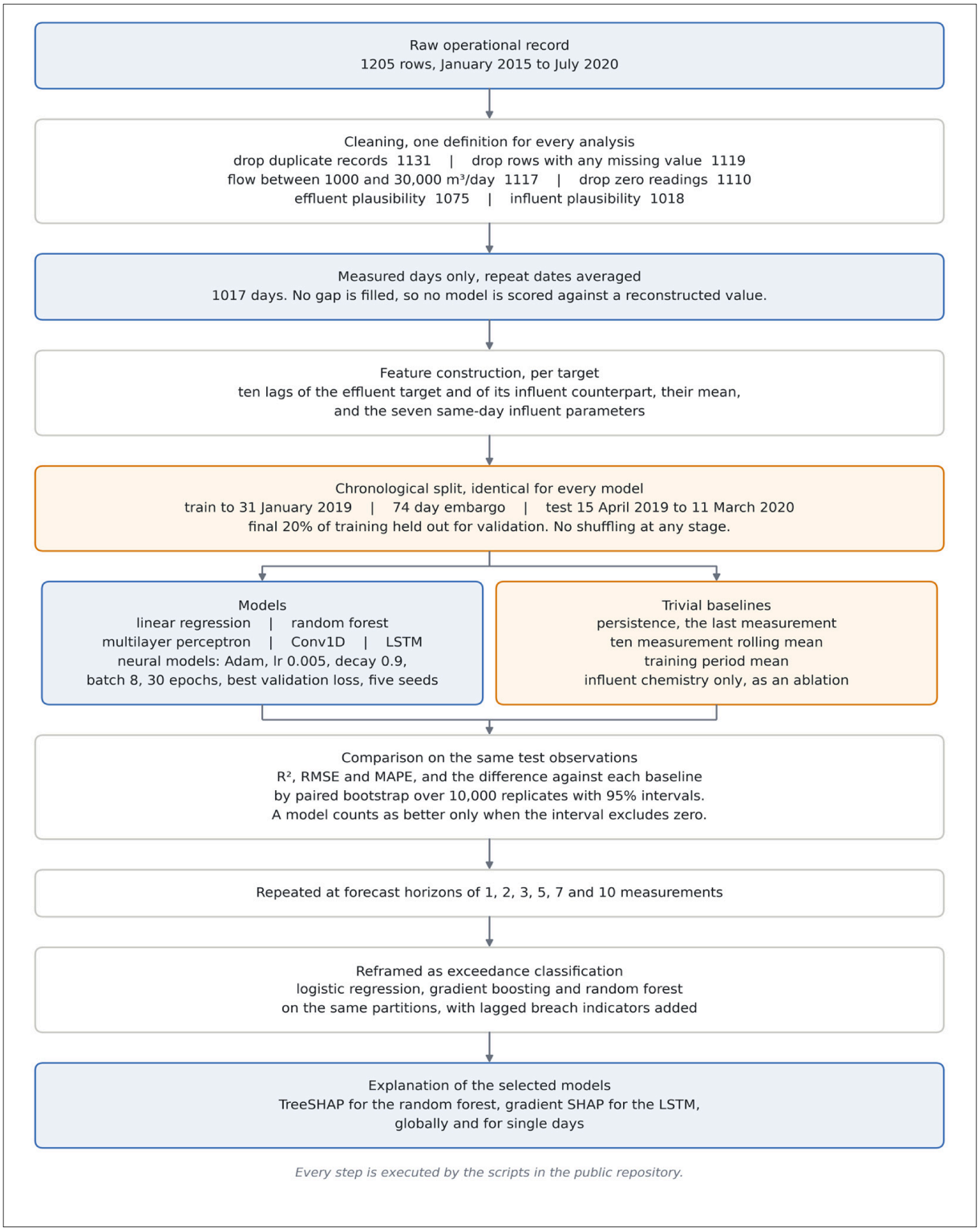


Figure 1. The modeling pipeline, from the raw operational record to the explanation of the selected models. Record counts at each cleaning step are those produced by the analysis code.

2.5. Data Partitioning

A single chronological partition was applied to all models. The training period runs from 22 January 2015 to 31 January 2019 (812 records) and the held-out test period from 15 April 2019 to 11 March 2020 (150 records), separated by a 74-day embargo. The embargo ensures that no lag feature of any test observation can draw on data used for training. Within the training period the final 20% of records was reserved for validation and model selection, again chronologically and with a further gap of one lag window. No random shuffling was applied at any stage. All models, including linear regression (LR), were trained and evaluated on exactly these partitions and on identical observations. For the robustness analysis reported in Section 3.4, two additional chronological splits were used: training to the end of 2016 with 2017 as test, and training to the end of 2017 with 2018 as test, each with the same embargo construction.

2.6. Models, Baselines and Statistical Testing

Four model families were evaluated for regression. LR was fitted by ordinary least squares on the full predictor set. A multilayer perceptron (MLP) received the lagged variables as flat features through a single hidden layer of 200 units with Rectified Linear Unit (ReLU) activation. A one-dimensional convolutional network (Conv1D) received the lagged variables as a sequence of shape (10, 2) through a single convolutional layer with 3 output channels and kernel size 5. LSTM network received the same sequence representation through a single recurrent layer with hidden state dimension 10, taking the final hidden state of the sequence. In the convolutional and recurrent models, the sequence representation is concatenated with the same-day influent features before the shared fully connected layer.

All neural models were trained with the Adam optimiser at an initial learning rate of 0.005 with exponential decay (gamma = 0.9), batch size 8, mean squared error loss, for 30 epochs, retaining the parameter set achieving the lowest validation loss. Each configuration was trained with five fixed random seeds and we report the mean and standard deviation (SD) across seeds. For exceedance classification, logistic regression, gradient boosting and random forest classifiers were evaluated on the same partitions, using lagged concentrations and lagged exceedance indicators as predictors.

A random forest regressor (300 trees, minimum leaf size 3) and a ridge regression were added to the comparison, since the reviewers asked for a broader and more rigorous benchmark than the original submission provided. Two reference predictors are reported for every model and every horizon: persistence that is the most recent observed value, and the mean of the ten preceding measurements. Reporting both matters because neither dominates; persistence is the stronger reference at short horizons and the rolling mean at long ones, so a model must beat both before any predictive gain can be claimed. The training-period mean is also reported. For classification the corresponding baselines were the previous exceedance indicator, the rolling exceedance rate and the rolling concentration. An influent-only model, using the seven same-day influent parameters without any lagged effluent term, was fitted as an ablation. Differences in R^2 and area under the curve (AUC) between each model and its baseline were assessed by paired bootstrap resampling of the test observations over 10,000 replicates, with 95% percentile intervals reported. Seasonal differences in removal efficiency were tested with the Mann–Whitney U test.

Models were additionally evaluated at forecast horizons of 1, 2, 3, 5, 7 and 10 measurement steps. At each horizon the model predicts the effluent concentration h steps ahead using only information available at the time of prediction, namely influent chemistry, the ten preceding effluent and influent measurements, their rolling mean, and the time of year. The baselines use the identical information set. Given the observed sampling interval, a horizon of five measurements corresponds to approximately ten days and a horizon of ten to approximately three weeks. This protocol was adopted because the one-step-ahead task, which dominates the published literature, is the regime in which trivial extrapolation is hardest to beat and is also of limited operational value.

3. Results and Discussion

3.1. Descriptive Statistics

Table 1 reports descriptive statistics for the influent and effluent parameters and the operating variables over the full monitoring period. Flow records outside the range 1000 to 30,000 m^3/day were removed before computing these statistics, as a small number of erroneous entries propagate into the derived HRT. After this correction the HRT has a mean of 0.9 days and a maximum of 3.2 days, consistent with the plant configuration. Flow, HRT, excess sludge and biogas production are reported for plant characterisation; of these only flow enters the models, through the computation of mass loads.

Mean removal efficiencies over the monitoring period were 95.6% for COD, 91.6% for BOD_5, 77.8% for TN, 80.3% for TP and 94.8% for TSS. Influent composition is highly variable, with the coefficient of variation reaching 0.74 for suspended solids, while effluent concentrations are considerably more stable in absolute terms. This contrast is quantified in Section 3.6.

Table 1. Descriptive statistics of influent, effluent and operating variables over the monitoring period, after removal of physically implausible flow records.

	Mean	Std	Min	25%	50%	75%	Max
Temperature (in) [°C]	18.6	3.6	9.0	15.5	18.7	21.7	26.8
pH (in)	7.6	0.8	4.9	7.1	7.7	8.1	9.4
COD [mg/L] (in)	960.8	301.8	10.7	743.0	1100.0	1155.0	1884.0
BOD_5 [mg/L] (in)	422.8	149.3	54.0	321.0	411.0	519.0	820.0
N [mg/L] (in)	77.4	25.0	5.2	62.1	73.9	88.8	190.0
P [mg/L] (in)	13.5	5.4	2.8	9.8	12.1	15.9	33.8
TSS [mg/L] (in)	1048.9	775.0	2.4	496.0	874.0	1425.0	4866.0
COD [mg/L] (out)	42.7	20.2	1.7	30.2	38.0	49.0	121.0
BOD_5 [mg/L] (out)	35.3	32.0	0.0	14.1	25.1	47.8	172.0
TN [mg/L] (out)	17.2	11.9	0.0	8.7	13.5	20.6	69.9
TP [mg/L] (out)	2.7	2.3	0.1	1.1	2.1	3.7	38.6
TSS [mg/L] (out)	54.1	85.1	0.0	10.0	26.0	66.0	818.0
Q [m^3/day]	9735.9	2238.4	2718.0	8493.0	9344.0	10,395.0	22,712.0
HRT [days]	0.9	0.2	0.4	0.8	0.9	1.0	3.2
Excess sludge [m^3]	497.7	146.9	96.0	385.0	595.0	596.0	1316.0
Biogas [m^3/day]	1867.3	498.8	0.0	1629.0	1886.0	2157.8	3287.0

Figure 2 shows influent and effluent concentrations through time for the monitored parameters. The influent series are noisy, with pronounced short-term excursions, while the effluent series are visibly attenuated.

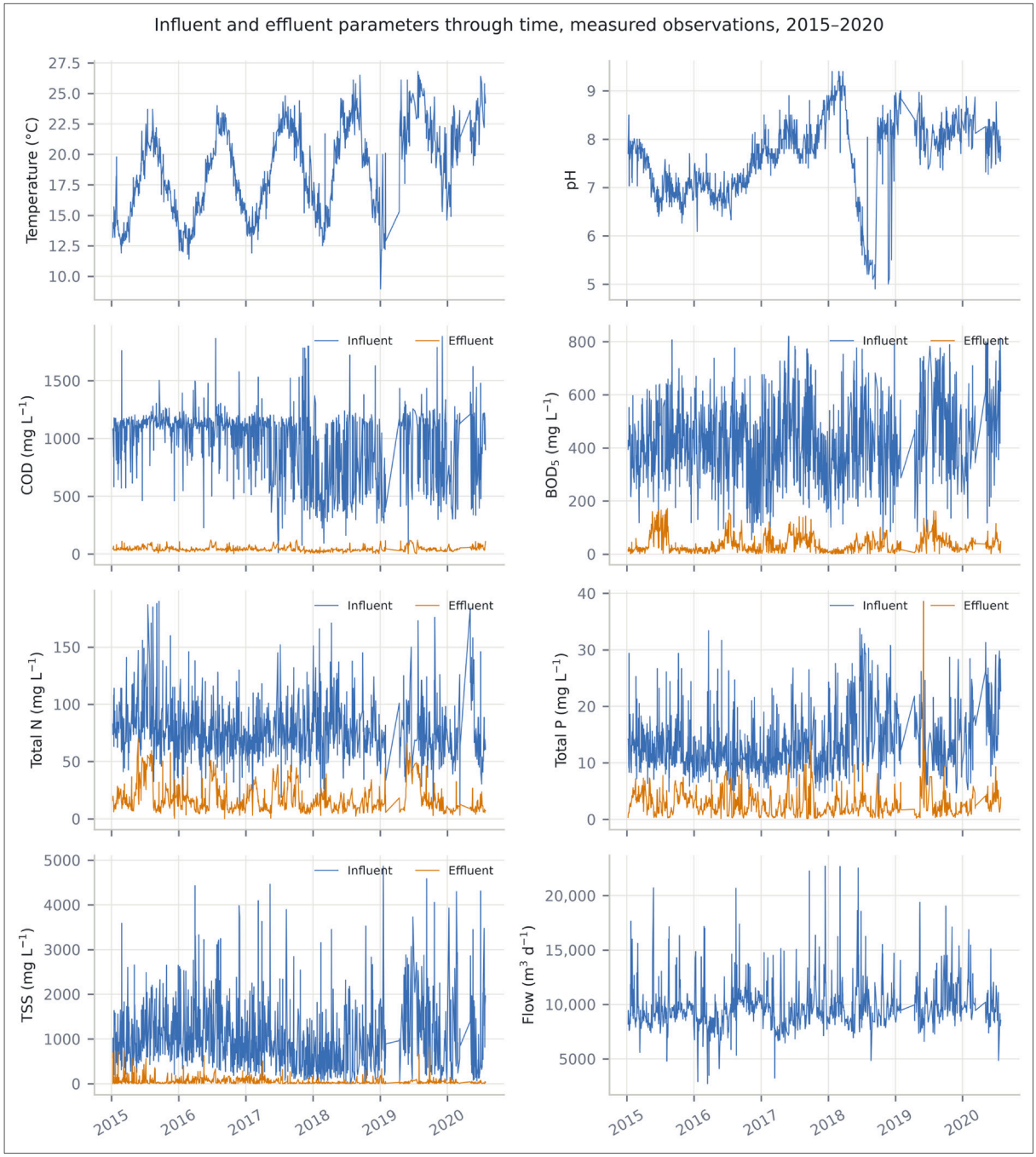


Figure 2. Influent and effluent concentrations of COD, BOD_5, TN, TP and TSS, together with influent temperature and pH, over the monitoring period January 2015 to July 2020 (measured observations only).

3.2. Correlation Matrix

Pearson correlations between the influent parameters, the lagged effluent terms and the effluent target were computed on measured observations only. Following the observation of Reviewer 1 regarding collinearity, the individual lagged columns are omitted from the display in favour of the immediately preceding measurement and the ten-measurement rolling mean.

Figure 3 shows the correlation matrix for effluent COD and Figure 4 for effluent BOD_5.

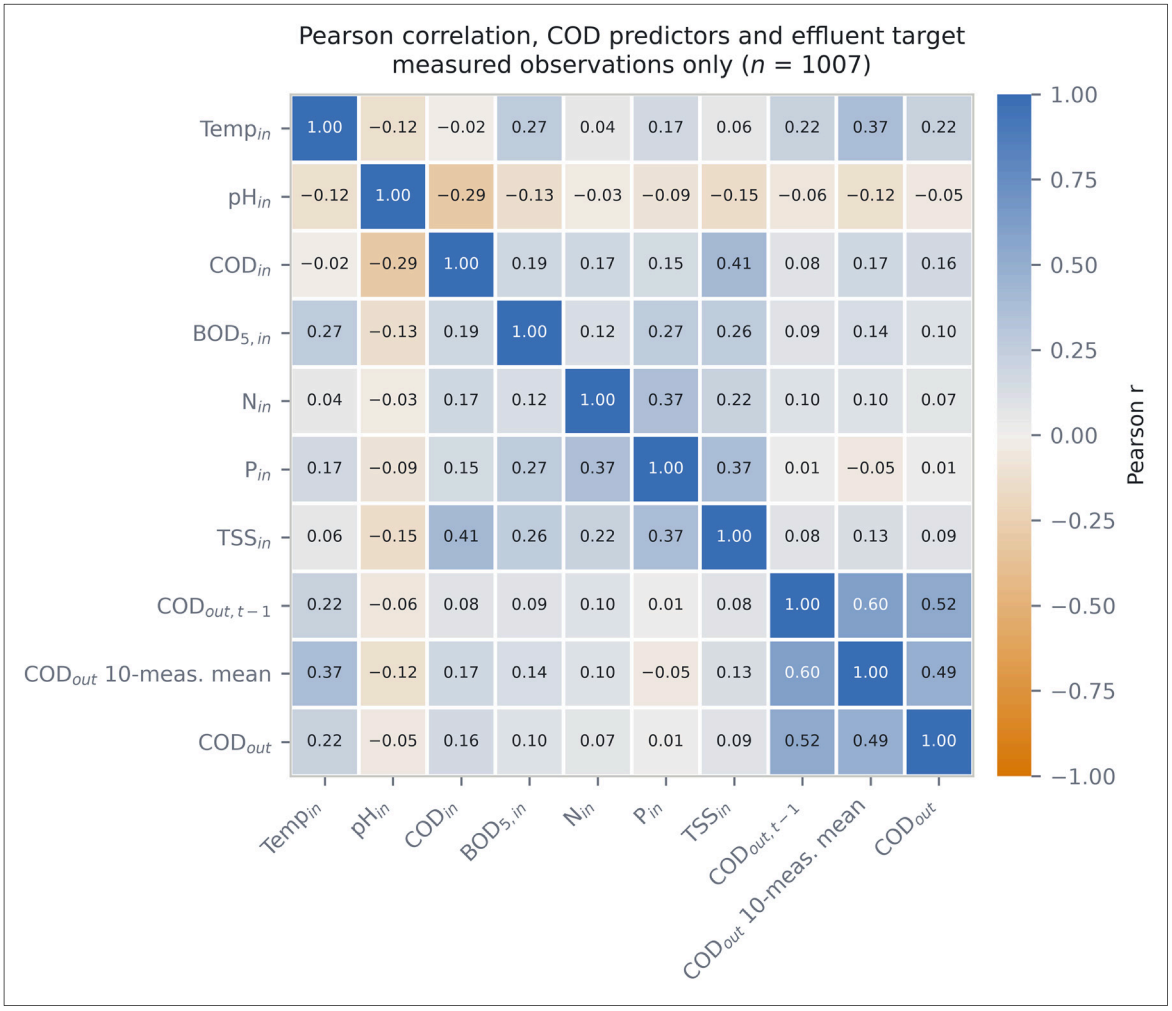


Figure 3. Pearson correlation between influent parameters, lagged effluent terms and effluent COD, measured observations only (n = 1007). Statistical significance was assessed at $p < 0.05$.

The influent parameters show weak association with effluent quality. For effluent COD the correlations are +0.22 with influent temperature, +0.16 with influent COD, +0.10 with influent BOD_5 and below +0.10 for pH, nitrogen, phosphorus and suspended solids. For effluent BOD_5 the association with influent temperature is stronger at +0.40, which anticipates the seasonal result reported in Section 3.5. These weak associations are not incidental; they are a central empirical observation of this study and are explained in Section 3.6.

By contrast, the effluent series correlates strongly with its own recent history: +0.52 between effluent COD and the immediately preceding measurement, and +0.49 with the ten-measurement rolling mean. For BOD_5 the equivalent values are +0.62 and +0.65. Effluent quality at this plant is therefore governed far more by process inertia than by measurable day-to-day variation in influent composition.

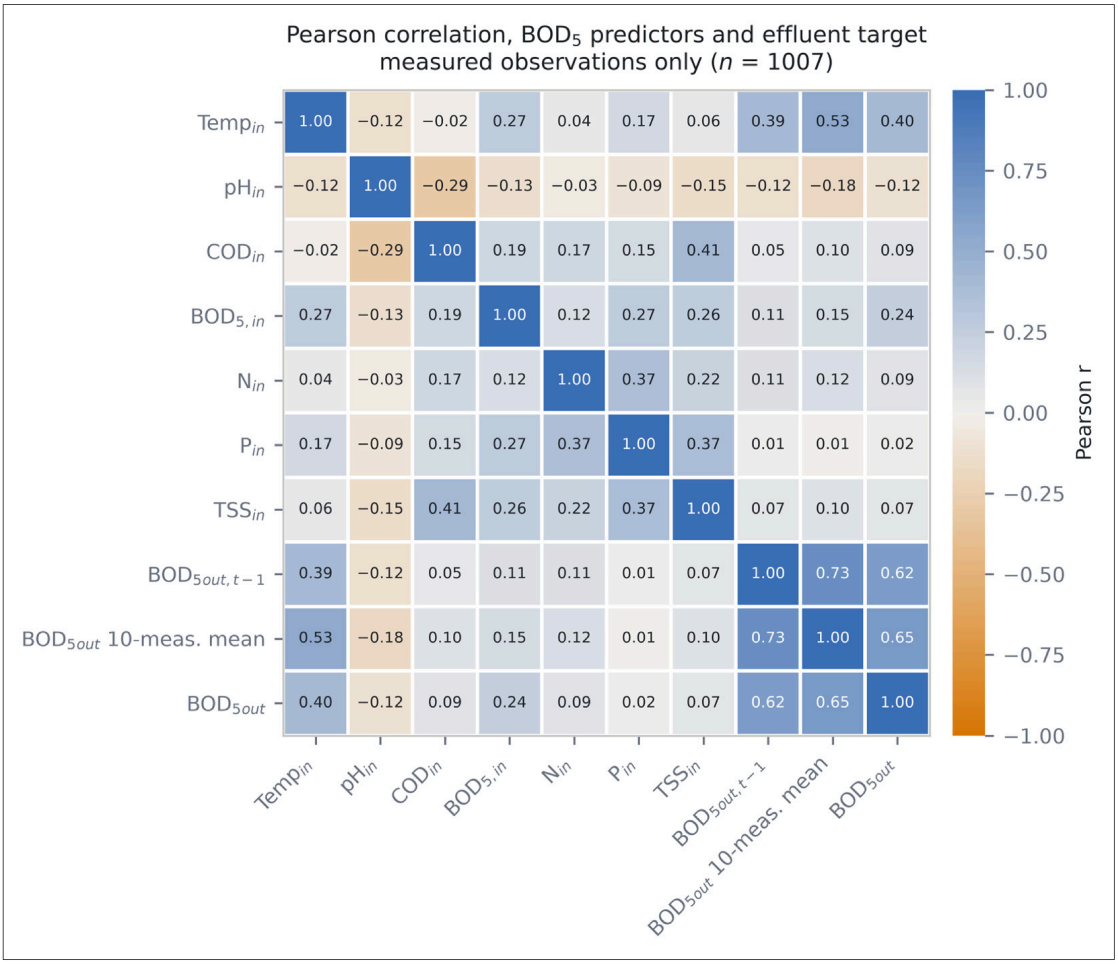


Figure 4. Pearson correlation between influent parameters, lagged effluent terms and effluent BOD5, measured observations only (n = 1007). Statistical significance was assessed at $p < 0.05$.

3.3. Violin Plots

Violin plots summarise the distribution of each parameter, including skewness, multimodality and the extent of the tails, and were used to guide the plausibility filters applied in Section 2.4. No distributional transformations were applied to the data (Figure 5).

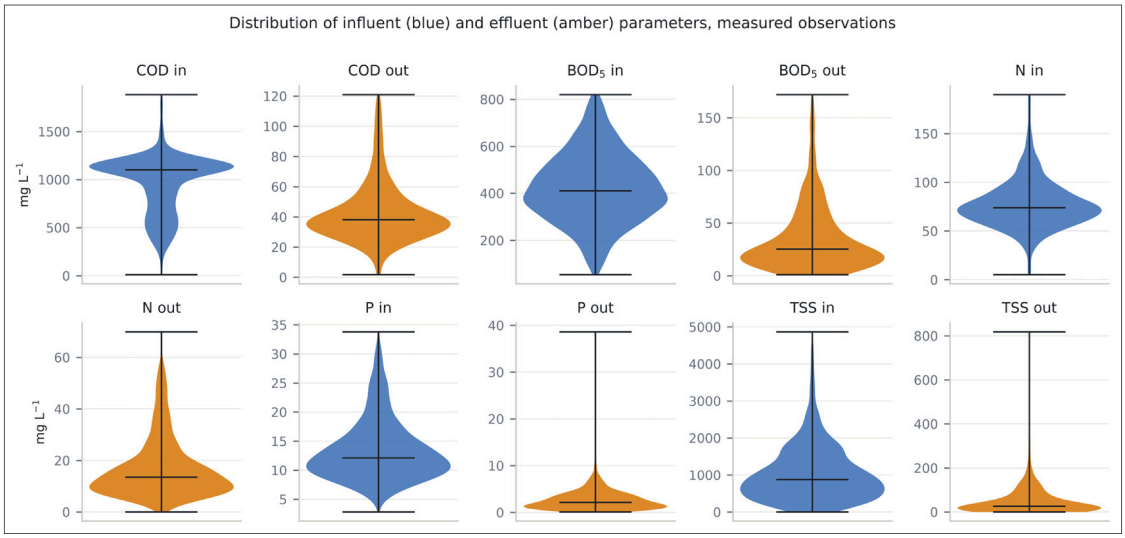


Figure 5. Violin plots of the distribution of each influent and effluent parameter over the monitoring period, measured observations only.

Figure 5 shows that the influent distributions are strongly right-skewed with long upper tails, while the effluent distributions are narrower in absolute terms. This contrast is quantified in Section 3.6.

3.4. Predictive Performance for Effluent Concentration

Four model families were evaluated: LR, a multilayer perceptron, a one-dimensional convolutional network and a LSTM network. All were trained and evaluated on the shared chronological partition described in Section 2.5, on identical observations, and compared against three naive baselines and an influent-only ablation.

Figure 6 shows measured against predicted effluent concentrations over the held-out test period for the linear model. Predictions cluster around the identity line at moderate concentrations but systematically under-predict the upper tail, which is the operationally important region.

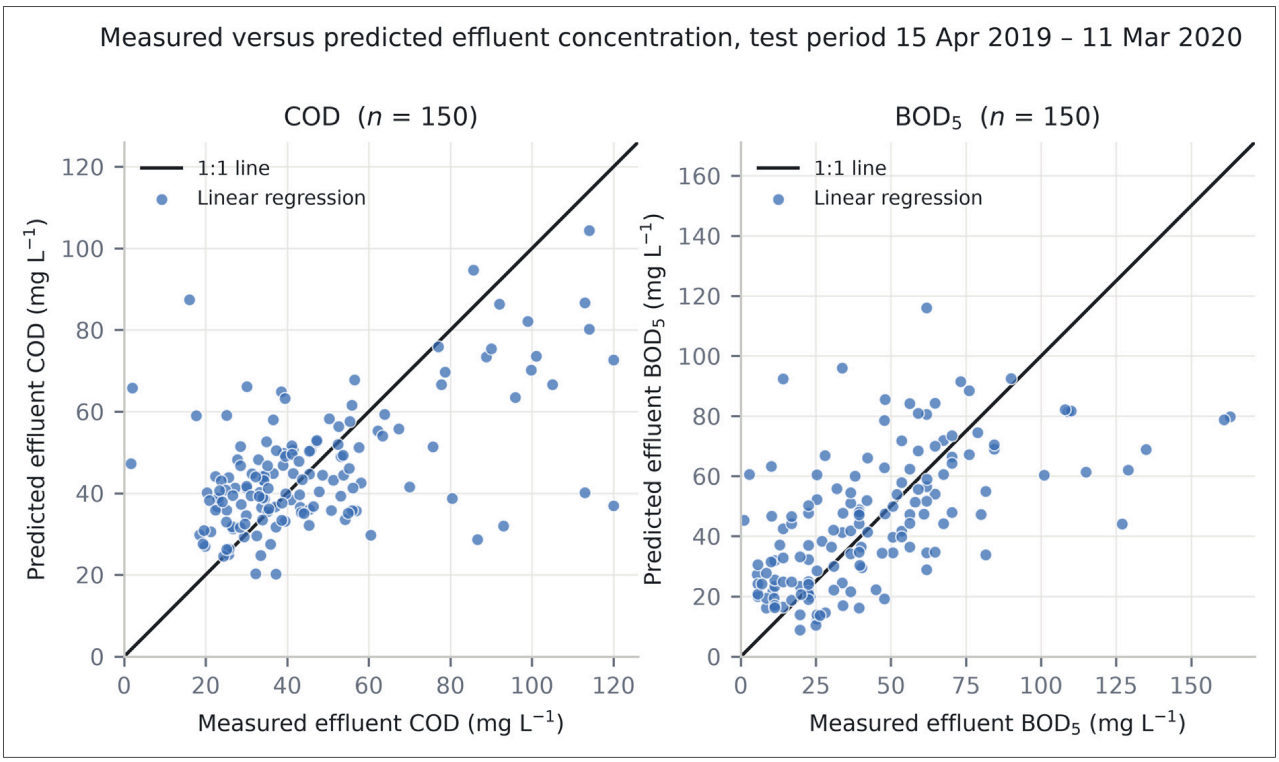


Figure 6. Measured against predicted effluent COD and BOD_5 for the LR model over the held-out test period 15 April 2019 to 11 March 2020 (n = 150 measured observations for each target).

Tables 2 and 3 report test-set performance for effluent COD and BOD_5 using the root mean square error (RMSE), mean absolute percentage error (MAPE), and R^2. Values for the neural models are means over five random seeds with SDs. All models are scored on the same observations.

The LSTM is the strongest model for both targets and by far the most stable, with a SD across seeds of 0.004 against 0.144 for the multilayer perceptron on COD. However, no model outperforms the ten-measurement moving average by a statistically significant margin. Paired bootstrap over 10,000 replicates gives a difference of −0.007 for COD (95% confidence interval (CI) −0.063 to +0.078) and +0.028 for BOD_5 (95% CI −0.025 to +0.105). Both intervals contain zero, and for COD the baseline nominally exceeds every model tested.

Table 2. Predictive performance for effluent COD over the held-out test period (n = 150 measured observations). Baselines and the ablation are shown in italics.

Method	R^2	RMSE [mg/L]	MAPE
LSTM	0.321 ± 0.004	20.36	0.806
LR	0.308	20.55	0.732
MLP	0.195 ± 0.144	22.10	0.806
Conv1D	0.123 ± 0.073	23.12	0.815
Baseline: ten-measurement mean	0.328	20.26	0.882
Baseline: previous measurement	0.063	23.91	0.686
Baseline: training-period mean	−0.035	25.13	0.664
Ablation: influent chemistry only	0.051	24.07	0.689

Table 3. Predictive performance for effluent BOD_5 over the held-out test period (n = 150 measured observations). Baselines and the ablation are shown in italics.

Method	R^2	RMSE [mg/L]	MAPE
LSTM	0.380 ± 0.002	24.52	1.185
LR	0.348	25.14	1.091
MLP	0.345 ± 0.011	25.20	1.080
Conv1D	0.259 ± 0.041	26.80	1.126
Baseline: ten-measurement mean	0.352	25.07	1.269
Baseline: previous measurement	0.017	30.87	1.168
Baseline: training-period mean	−0.100	32.65	1.098
Ablation: influent chemistry only	0.207	27.73	1.423

The influent-only ablation reaches R^2 of 0.051 for COD and 0.207 for BOD_5, confirming that influent chemistry as routinely measured carries little information about same-day effluent quality. Extending the analysis to all five effluent parameters shows that this is a property of the parameter rather than of the modeling approach. Effluent total nitrogen reaches R^2 of 0.740 against a baseline of 0.600, a difference of +0.140 (95% CI +0.041 to +0.250), and is the only parameter for which the model exceeds the baseline (Figure 7) on all three chronological splits tested (+0.349 for the 2017 test period, +0.321 for 2018 and +0.140 for 2019 to 2020). COD is significant on none of the three splits and BOD5 on one, which is consistent with noise. We note that the nitrogen advantage derives from the lag structure rather than from influent chemistry, an influent-only model reaching R^2 of 0.131 against 0.740 for the full specification. Effluent nitrogen is therefore more usefully autocorrelated than COD or BOD_5, rather than being driven by measurable influent variation.

Two qualifications apply to Tables 2 and 3. First, the ten-measurement mean is not the only trivial alternative, and it is not the strongest one for every parameter. Against persistence, that is simply the most recent observation, the ordering reverses: the model exceeds persistence significantly for COD and BOD_5 but not for total nitrogen. At a horizon of one measurement, therefore, no parameter is predicted significantly better than both trivial baselines. Second, and consequently, the one-step-ahead task is a poor basis on which to judge whether machine learning is useful. We therefore examine the dependence on forecast horizon. This finding provides important context for comparison with previous studies (Table 4), which generally report higher R^2 values.

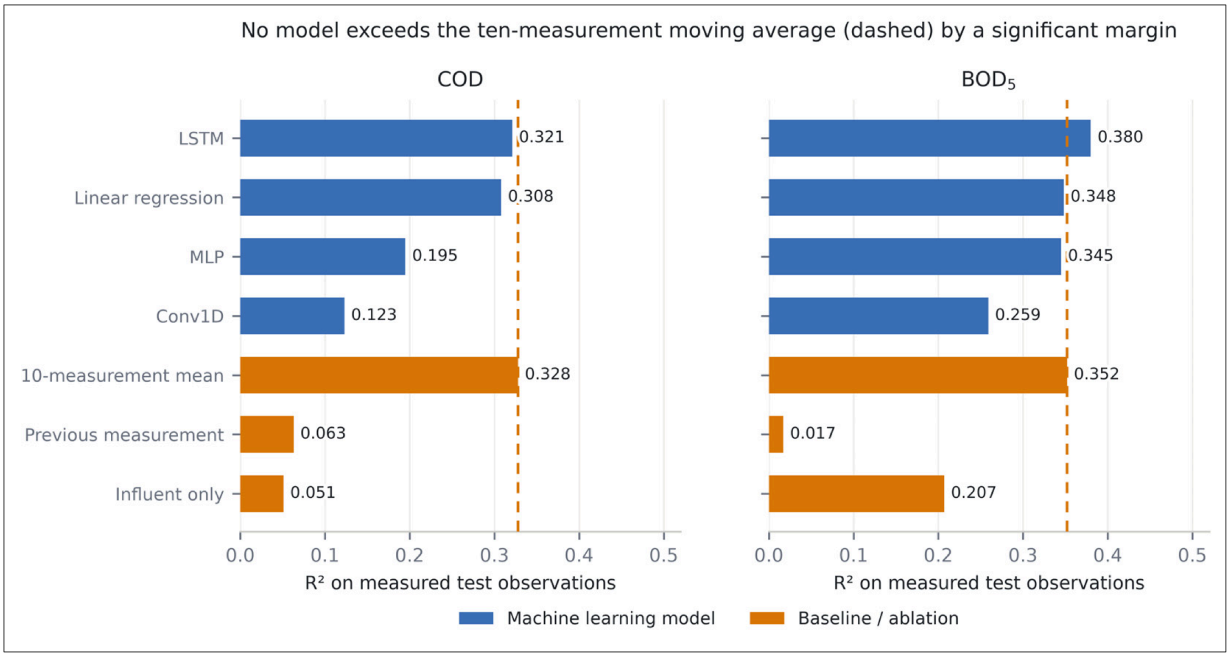


Figure 7. Predictive performance by effluent parameter against the ten-measurement moving-average baseline, held-out test period, measured observations only. The dashed vertical line marks the coefficient of determination of that baseline.

Table 4. Comparison of ML-based effluent quality prediction performance in other studies.

Effluent Parameter	ML Method	Reported R^2	Reference
COD	LSTM	0.634	[27]
	MLP	0.487	[27]
	MLP	0.87	[28]
	LSTM	0.55	[28]
	LSTM	0.93	[29]
	LSTM	0.727	[30]
BOD_5	LR	0.40	[31]
Total nitrogen	LSTM	0.876	[27]
	MLP	0.697	[27]
	LSTM	0.76	[29]

Such comparisons should be interpreted cautiously, as previous studies often rely on shorter monitoring periods and less stringent benchmarking frameworks. The six-year operational dataset used here contains greater seasonal and process variability, while the common evaluation framework incorporates explicit naive baselines and multiple forecasting horizons. The lower R^2 values observed in the present study may reflect a more challenging and rigorous assessment of predictive performance rather than inferior model capability.

Dependence on Forecast Horizon

Trivial extrapolation carries no information beyond the most recent observations and must therefore decay as the forecast horizon lengthens. A model that also uses influent chemistry and seasonal structure need not decay at the same rate. Table 5 reports performance at horizons of one, five and ten measurement steps, corresponding to approximately two days, ten days and three weeks of plant operation. Figure 8 shows the same dependence across the full set of horizons tested.

Table 5. Coefficient of determination on the held-out test period as a function of forecast horizon (n = 150 at every horizon). Neural results are means over three seeds with SDs.

Target	Horizon	Random Forest	LSTM	Persistence	Rolling Mean
TN	1	0.745	0.599 ± 0.007	0.679	0.601
	5	0.516	0.317 ± 0.070	0.388	0.320
	10	0.272	0.147 ± 0.005	0.078	0.001
BOD5	1	0.338	0.363 ± 0.007	0.008	0.349
	5	0.294	0.299 ± 0.029	−0.246	0.245
	10	0.204	0.215 ± 0.003	−0.349	0.016
COD	1	0.361	0.323 ± 0.005	0.064	0.327
	5	0.186	0.156 ± 0.038	−0.261	0.202
	10	0.071	−0.038 ± 0.026	−0.231	−0.048

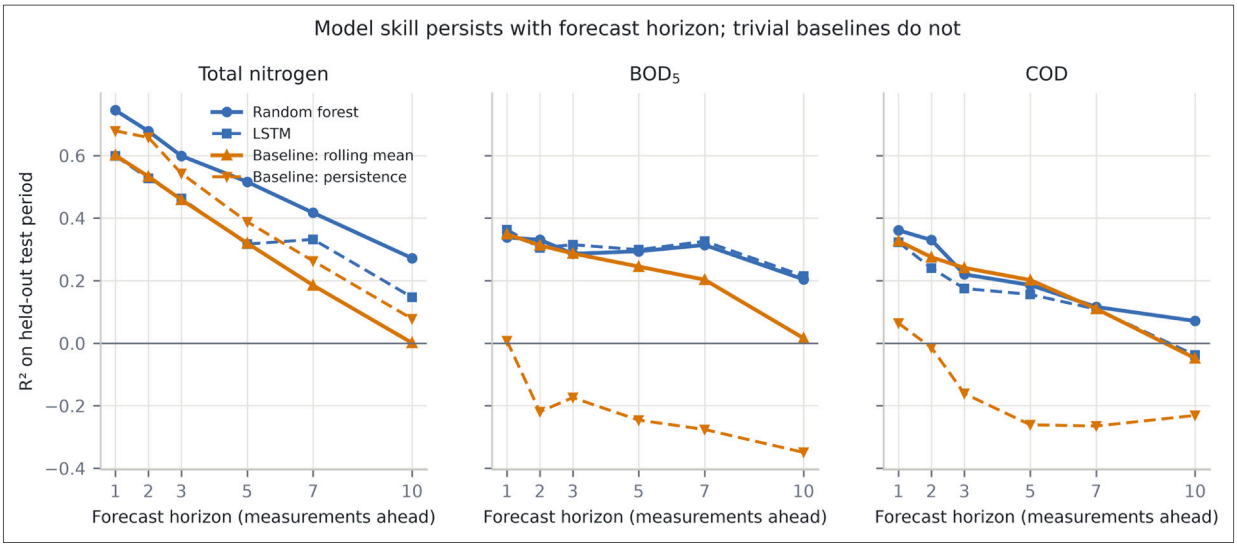


Figure 8. Predictive skill against forecast horizon for the two model families and the two trivial baselines. Baseline skill decays to zero or below; model skill decays more slowly.

The pattern is consistent across parameters. Baseline skill falls rapidly: for total nitrogen the rolling mean declines from R^2 of 0.601 at one step to 0.001 at ten steps, and persistence from 0.679 to 0.078. Model skill declines far more slowly, the random forest retaining 0.272 at ten steps. For BOD_5 persistence becomes actively harmful beyond one step, falling to −0.349, while the recurrent model remains 0.215.

Two results exceed both trivial baselines at conventional significance. For total nitrogen at a horizon of five measurements the random forest reaches R^2 of 0.516 against 0.388 for persistence (95% CI on the difference +0.009 to +0.288) and 0.320 for the rolling mean (+0.076 to +0.336). For BOD_5 at a horizon of ten measurements both the random forest and the LSTM exceed both baselines, the LSTM reaching 0.215 against −0.349 and 0.016 respectively (+0.215 to +0.994 and +0.021 to +0.430). COD is the exception: the random forest exceeds persistence at short horizons but never exceeds the rolling mean at any horizon, consistent with the attenuation reported in Section 3.6.

This bears directly on how predictive performance should be reported. The one-step-ahead task that dominates the published literature is precisely the regime in which trivial extrapolation is most competitive, and it is also the regime of least operational value, since a warning issued one sampling interval in advance leaves no time to intervene. It also resolves an apparent contradiction in our own results: LR is competitive with, and at one step ahead marginally better than, the more complex architectures, but it degrades faster, falling to R^2 of 0.059 for BOD_5 at ten steps where the LSTM retains 0.215. The value

of the sequence model is not visible at one step ahead and becomes clear at operational lead times.

3.5. Seasonal Deterioration of Biological Treatment

Aggregating six years of records by month reveals a systematic pattern that had not previously been examined at this plant. Between June and August, total nitrogen removal falls from 79.4% to 67.4% and mean effluent BOD_5 rises from 26.7 to 62.2 mg/L. Table 6 summarises the comparison and Figure 9 shows the monthly structure.

Table 6. Treatment performance in summer against the remainder of the year, 2015 to 2020. Significance by Mann–Whitney U test.

	Summer (June to August)	Rest of Year	Difference
TN removal	67.4%	79.4%	−12.0 pp ($p = 4 \times 10^{-18}$)
BOD_5 removal	85.5%	92.8%	−7.3 pp ($p = 1 \times 10^{-34}$)
COD removal	91.2%	95.4%	−4.2 pp ($p = 5 \times 10^{-13}$)
TSS removal	92.8%	92.1%	+0.7 pp (not significant)
Effluent total nitrogen	24.9 mg/L	14.7 mg/L	+10.2 mg/L
Effluent BOD_5	62.2 mg/L	26.7 mg/L	+35.5 mg/L ($p = 6 \times 10^{-44}$)
Effluent COD	53.9 mg/L	39.2 mg/L	+14.7 mg/L ($p = 1 \times 10^{-20}$)
TP removal	78.3%	78.4%	−0.1 pp (not significant)
Influent total nitrogen	79.0 mg/L	76.9 mg/L	+2.1 mg/L
Hydraulic flow	9710 m^3/day	9749 m^3/day	−39 m^3/day

The pattern is present in five of the six monitored years. Nitrogen removal in summer against the rest of the year was 56.9% against 78.0% in 2015, 66.8% against 81.6% in 2016, 67.8% against 77.2% in 2017, 81.9% against 79.6% in 2018, 54.8% against 77.6% in 2019 and 81.9% against 86.9% in 2020. The year 2018 is an exception and is reported as such.

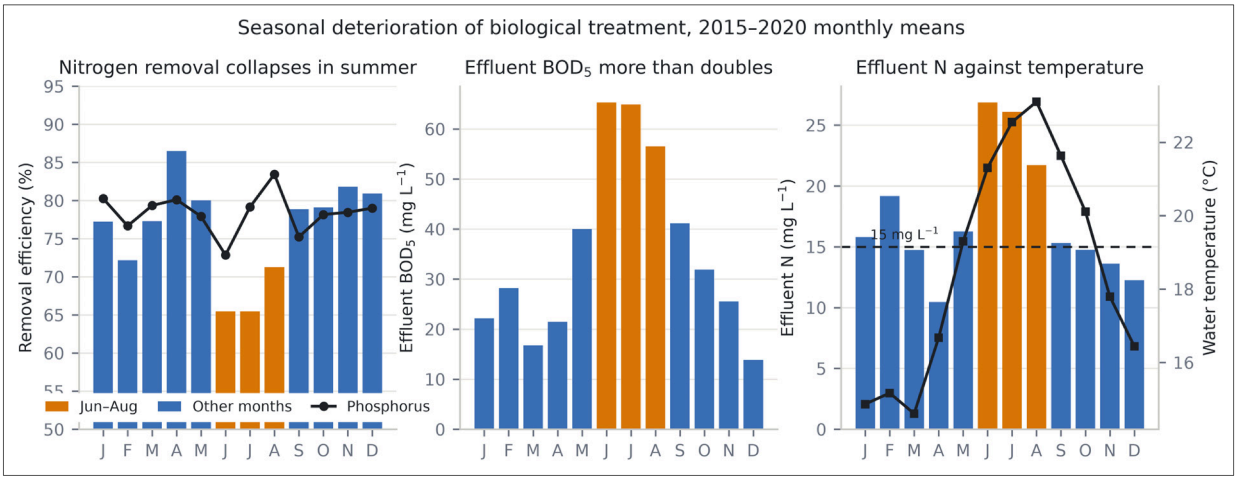


Figure 9. Monthly means 2015 to 2020. Left: nitrogen removal efficiency, with phosphorus removal shown for comparison. Centre: effluent BOD_5. Right: effluent total nitrogen against water temperature, with the 15 mg/L regulatory threshold indicated. June to August is highlighted.

Three observations constrain the interpretation. First, the effect is not attributable to loading: influent total nitrogen differs by 2.1 mg/L between seasons and hydraulic flow by less than 0.4%. The plant receives an equivalent load and treats it less effectively. Second, the magnitude of the deficit orders exactly as oxygen demand does. Nitrification, the most oxygen-intensive step, is worst affected at −12.0 percentage points, followed by the oxidation of organic matter at −7.3 for BOD_5 and −4.2 for COD, while removal of

suspended solids, which proceeds by physical settling and requires no oxygen, is unaffected at +0.7 points. Phosphorus removal is likewise unchanged. The absence of any effect on suspended solids is important because it excludes the most obvious competing explanation, namely deterioration of sludge settling and carryover of solids into the effluent, which would raise all effluent concentrations together. Third, the deterioration coincides with the annual maximum in water temperature.

Taken together these observations point to oxygen limitation as the leading hypothesis. Dissolved oxygen saturation falls by approximately 20% between 15 °C and 23 °C, the range spanned between the winter and summer means at this plant, while BOD_5 rises with temperature. Aeration capacity is therefore the most plausible binding constraint during precisely the period when demand is highest. This runs counter to the common expectation that warmer conditions improve nitrification kinetics, and indicates that at this plant the oxygen supply constraint dominates the kinetic benefit. We note that this remains an inference: the dataset contains no dissolved oxygen or aeration energy records, and confirming the mechanism directly would require both.

According to Serbian regulations, the applicable effluent limit for total nitrogen for a WWTP of this capacity is 15 mg/L [32]. Against this threshold, 60.8% of summer days exceed, against 38.5% for the remainder of the year. Treated effluent discharges to the Danube–Tisa–Danube canal, where eutrophication risk is greatest in exactly these months, and the revised Urban Wastewater Treatment Directive (UWWTD) (2024) tightens nitrogen requirements for plants of this scale [33].

3.6. Attenuation of Influent Variability

The weak association between influent and effluent quality reported in Section 3.2 has a physical explanation. Table 7 compares the SD of each parameter in the influent and in the effluent over the full monitoring period.

Table 7. Transfer of variability from influent to effluent over the full monitoring period, measured observations only.

Parameter	SD Influent [mg/L]	SD Effluent [mg/L]	Attenuation Factor	r (in, out)	Mean Removal
COD	301.0	20.2	14.9	+0.16	95.6%
TSS	775.9	85.2	9.1	+0.11	94.8%
BOD5	148.9	32.1	4.6	+0.24	91.6%
P	5.4	2.3	2.3	+0.11	80.3%
N	25.0	11.9	2.1	+0.21	77.8%

The ordering of attenuation matches the ordering of predictability reported in Section 3.4. Parameters that the plant buffers most strongly are those whose effluent concentration carries the least recoverable influent information, and total nitrogen, the least attenuated parameter, is the only one for which models exceed the baseline consistently. This provides a physical account of the negative result for COD and BOD_5: the limitation is not the modeling method but the information content of the effluent series at a plant with substantial buffering capacity. It also indicates that predictive gains at such plants should be sought for weakly buffered parameters rather than for the organic load indicators most commonly modeled in the literature.

3.7. Effect of Gap-Filling on Apparent Model Performance

Operational monitoring records are frequently incomplete, and it is common practice to resample to a regular interval and fill the gaps before modeling. Because our

record is sampled on approximately half of calendar days, we are able to quantify the consequence directly.

An earlier version of this analysis constructed a regular daily series by time-weighted interpolation followed by a fifty-day trailing rolling mean. Under that construction 49.3% of the modeled series consisted of reconstructed rather than measured values, and the held-out period was 54.5% reconstructed for COD and 49.7% for BOD_5. Scoring the identical model over the identical period separately on the two subsets gives the values in Table 8.

Table 8. LR performance on reconstructed against measured days, identical model and identical test period.

Test Subset	*n* (COD)	R^2 COD	*n* (BOD_5)	R^2 BOD_5
Reconstructed days only	181	0.817	165	0.913
Measured days only	151	0.340	167	0.397
Combined, as conventionally reported	332	0.486	332	0.576

Apparent performance on reconstructed days is more than double that on measured days. The reason is structural: interpolated and smoothed values are by construction close to a local average of their neighbours, and the strongest predictor available to the model is precisely such an average, so predicting a reconstructed value approaches circularity.

The effect extends beyond magnitude to model ranking. Under the gap-filled construction the LSTM was the weakest architecture, with R^2 of 0.387 for COD against 0.475 for the multilayer perceptron; on measured data it is the strongest. Gap-filling favours architectures that exploit local smoothness over those that model temporal structure, and can therefore reverse conclusions about which method is preferable. We do not suggest that interpolation is never appropriate. We do suggest that where a substantial fraction of an operational series has been reconstructed, performance should be reported on measured observations and the reconstructed fraction stated.

3.8. Exceedance Prediction and Operational Use

Regression asks what the effluent concentration will be. Plant operators face a different question, namely whether a regulatory threshold will be breached. Reframed as a classification task the models become operationally useful, as shown in Table 9.

Table 9. Exceedance prediction over the held-out test period 15 April 2019 to 11 March 2020. Precision and recall are reported at a decision threshold of 0.5.

Target	Best Model	AUC (95% CI)	Best Baseline AUC	Precision	Recall
TN above 15 mg/L	Random forest	0.896 (0.842–0.940)	0.847	0.80	0.71
BOD_5 above 25 mg/L	Logistic regression	0.825 (0.747–0.893)	0.813	0.86	0.88
TP above 2 mg/L	Random forest	0.767 (0.686–0.842)	0.717	0.67	0.66
COD above 50 mg/L	Random forest	0.805 (0.720–0.880)	0.809	0.81	0.46

For total nitrogen the model identifies 49 of 69 exceedance days with 12 false alarms at a decision threshold of 0.5. COD is the exception: the rolling-concentration baseline remains superior, which is consistent with the regression result and with the high attenuation reported in Section 3.6.

Removing all effluent history and supplying only influent chemistry and time of year provides an independent test of the seasonal finding in a predictive rather than descriptive setting. Under this restriction, nitrogen exceedance is predicted with an AUC of 0.713 (95% CI 0.628 to 0.793) and BOD_5 exceedance with 0.781 (0.701 to 0.855), both intervals lying well above 0.5. Seasonal harmonics carry 38% of the attribution for nitrogen and 44% for BOD_5, with influent temperature contributing a further 15% and 13%, respectively (Figure 10, right panel). We note that adding explicit seasonal terms to the full model does not improve it, since the lagged effluent terms already encode season implicitly; seasonal features matter when recent history is unavailable rather than in addition to it.

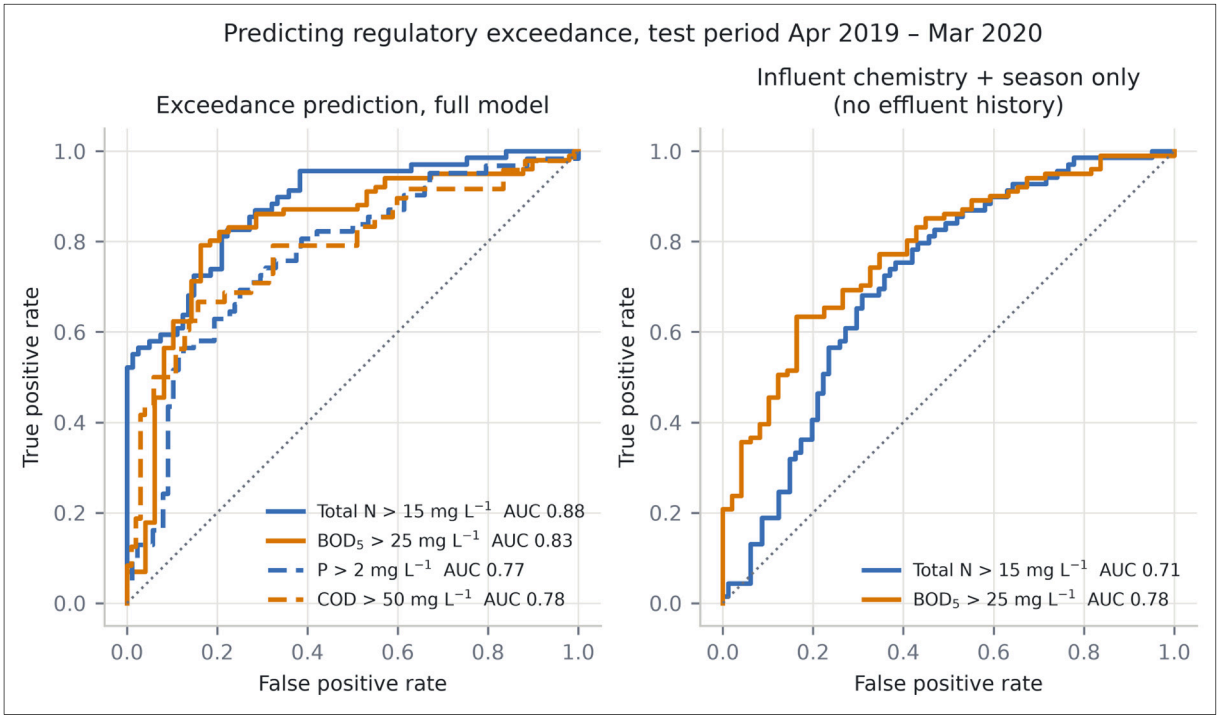


Figure 10. Receiver operating characteristic curves for exceedance prediction over the held-out test period. Left: full model using influent chemistry and recent effluent history. Right: model using influent chemistry and time of year only, with no effluent history. The dotted diagonal marks the performance of random guessing.

These results support a specific and modest deployment, and the horizon analysis in Section Dependence on Forecast Horizon determines its form. A one-step-ahead alarm has little operational value, because a warning issued one sampling interval in advance leaves no time to act, and it is also the regime in which trivial extrapolation is most competitive. Two configurations are defensible, and they are different tasks rather than one. As a forecast of concentration, total nitrogen at five measurements ahead reaches R^2 of 0.516 against 0.388 for persistence and 0.320 for the rolling mean; at ten measurements ahead the model retains 0.272 while the rolling mean falls to 0.001, so lead time can be bought at the cost of accuracy. As a threshold alarm, the same-day classifier of Table 9 reaches precision 0.80 and recall 0.71 at a decision probability of 0.5, which corresponds to approximately 61 alerts per year over the test period. These two sets of figures describe different tasks and should not be combined into a single claim, and Section Model Explanation and the Conditions Under Which the Alarm Is Useful sets out what the alarm can and cannot do. We deliberately do not specify aeration or coagulant dosing control logic, because the dataset contains no energy or chemical consumption records and any such specification would be unsupported by the evidence presented here.

Model Explanation and the Conditions Under Which the Alarm Is Useful

Impurity-based importance ranks predictors but carries no direction, is biased toward variables taking many distinct values, and cannot account for an individual day. Since the models proposed above are a random forest and a recurrent network, the explanation should match them. Shapley additive explanations (SHAP) were therefore computed for both: TreeSHAP, which is exact for tree ensembles, for the nitrogen exceedance classifier, and gradient SHAP for the network forecasting effluent BOD_5 ten measurements ahead.

Figure 11 shows the global attribution for the classifier. The dominant contributions are the previous effluent nitrogen measurement and the previous exceedance indicator, both raising predicted risk when high, followed by the second lag and the ten measurement mean. Influent temperature contributes little once recent effluent history is present, which is consistent with the observation in Section 3.8 that adding explicit seasonal terms does not improve the full model. Figure 12 shows the same analysis for the network, where influent solids, influent temperature and the seasonal harmonic carry most of the attribution at a ten measurement horizon, the regime in which recent effluent history has decayed in value.

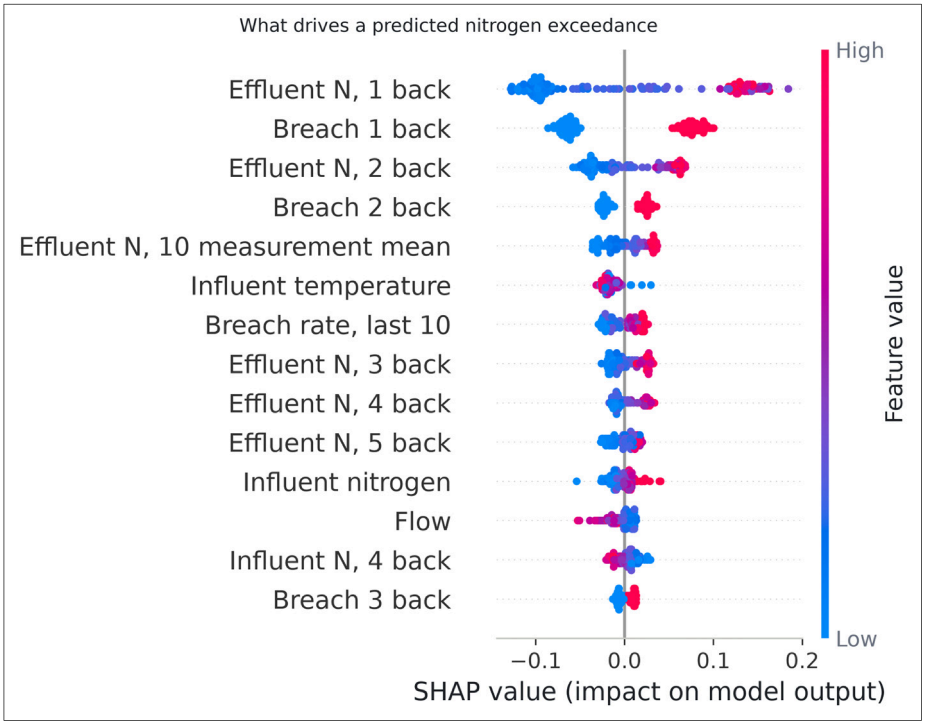


Figure 11. Shapley attribution for the random forest classifying total nitrogen exceedance above 15 mg/L over the held-out test period. Each point is one test day. Position gives the contribution to the predicted probability and colour gives the measured value of the variable.

Attribution for individual days is more informative than the global ranking, and it identifies a limitation that the aggregate performance figures conceal. Of the 69 exceedance days in the test period, 54 follow a measurement that was itself in exceedance and 15 follow a compliant measurement. At a decision probability of 0.5 the classifier identifies none of the latter, and the highest probability it assigns to any of them is 0.43. Figure 13 explains one such day. On 13 November 2019 effluent nitrogen reached 22.4 mg/L, and the largest single contribution to the model output was the absence of an exceedance at the preceding measurement, worth −0.06, against +0.04 for a preceding value of 14.7 mg/L that was already close to the limit. The reported precision of 0.80 and recall of 0.71 are therefore properties of a state monitor. The classifier recognises an exceedance episode that is already under way; it does not, at that threshold, announce one that is beginning.

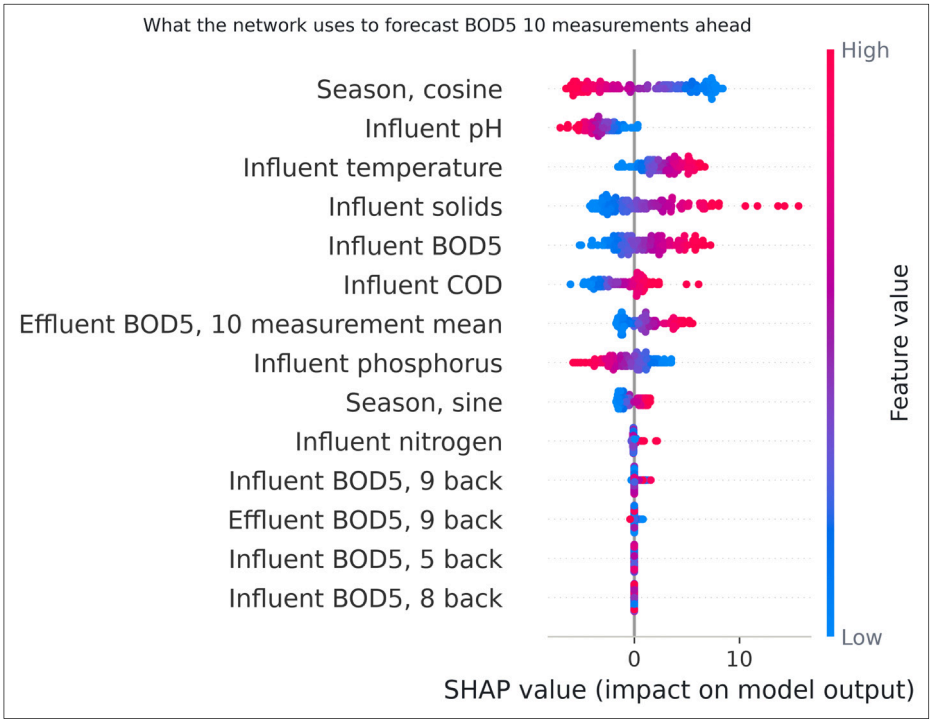


Figure 12. Shapley attribution for the LSTM network forecasting effluent BOD_5 ten measurements ahead over the held-out test period.

This is a property of the decision threshold rather than of the model's ranking. Restricted to the 81 test days that follow a compliant measurement, the same classifier separates the 15 exceedances with an AUC of 0.786 (95% CI 0.658 to 0.898), against 0.642 for the trivial alternative of ranking days by how close the preceding measurement already was to the limit. The information is present; 0.5 is simply the wrong operating point for a regime whose base rate is 19%. Figure 14 gives the trade-off. At a threshold of 0.35 the classifier identifies 47% of these exceedances at a precision of 0.54, which corresponds to approximately 16 alerts per 100 measurement days.

Two alternatives were examined and neither improved on this. A classifier trained only on days following a compliant measurement, with seasonal harmonics, removal efficiency, a short-term trend term and distance to the limit added to the predictor set, reached an AUC of 0.728 (0.579 to 0.854) when the model was selected on a validation period inside the training data, which does not improve on 0.642. Predicting an exceedance five or ten measurements ahead did not beat proximity to the limit at either horizon. We note that selecting the model on the test period instead returns 0.825 for the first of these, and report the validated figure.

The practical consequence is a two-threshold policy rather than a second model. A decision probability of 0.5 is appropriate while an exceedance episode is under way, where the classifier performs as reported in Table 9. A lower threshold of approximately 0.35 is appropriate when the preceding measurement was compliant, where it converts an alarm that never fires into one that identifies roughly half of beginning episodes at a precision close to three times the base rate. This distinction rests on 15 events in a single test period and should be confirmed on a longer record before it is relied upon operationally.

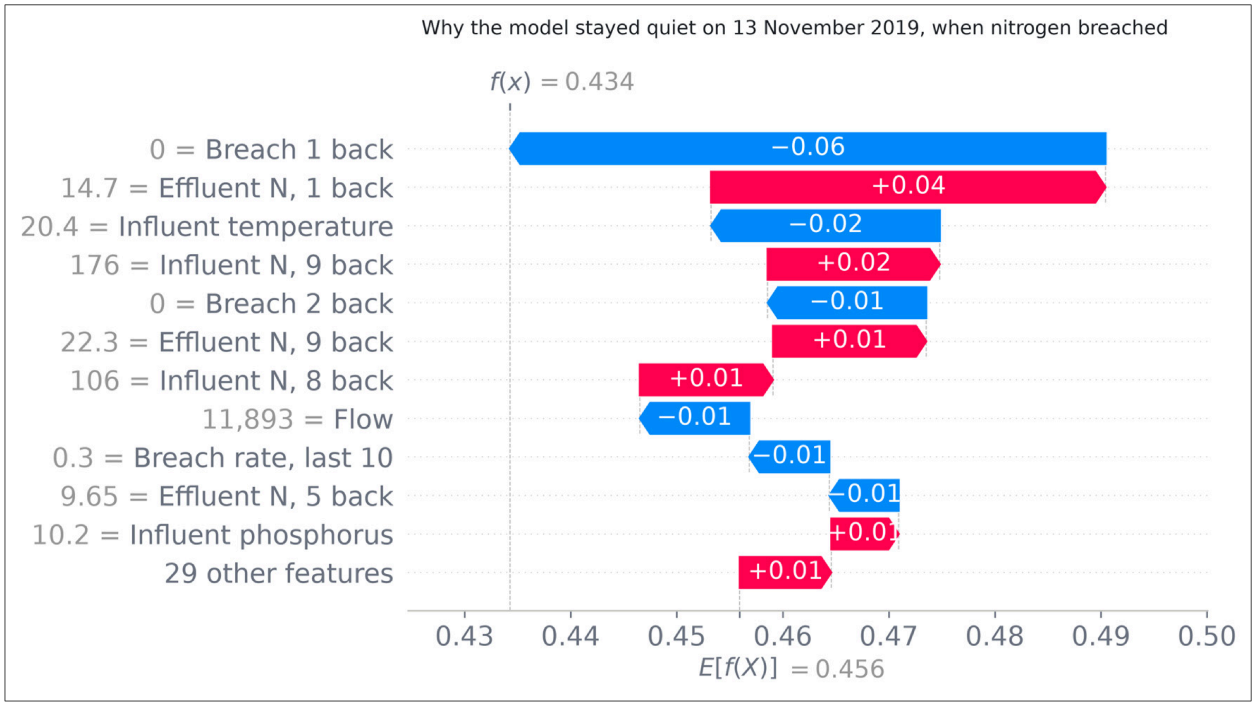


Figure 13. Attribution for 13 November 2019, an exceedance day preceded by a compliant measurement. The model output of 0.43 falls below the decision threshold of 0.5, and the absence of a preceding exceedance is the largest contributor to that outcome.

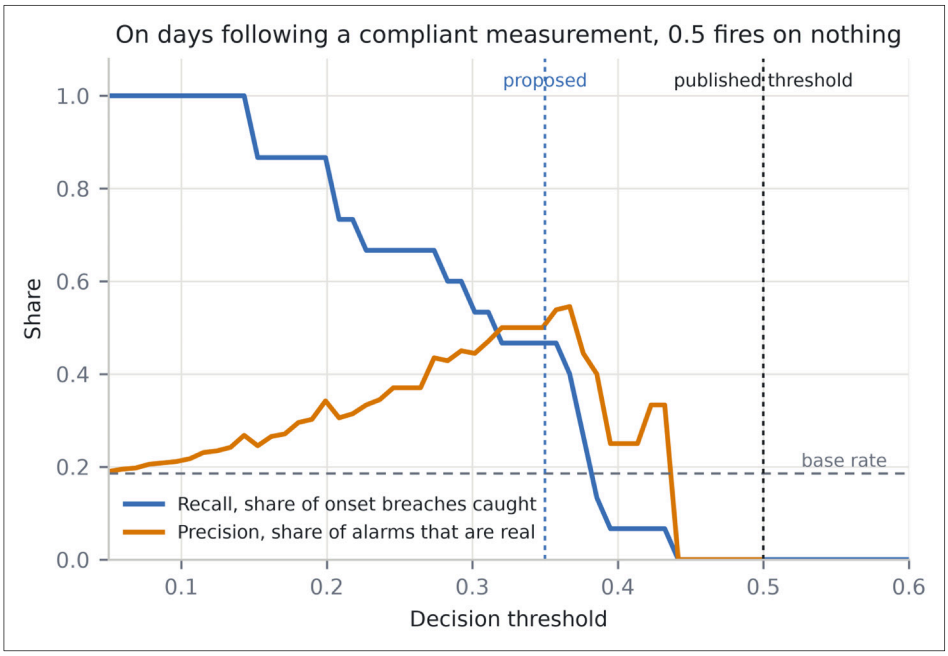


Figure 14. Recall and precision against decision threshold on the test days that follow a compliant measurement. The two curves cross near 0.35. Both are zero at the threshold of 0.5 used elsewhere in this paper.

3.9. Implications for Sustainable WWTP Management

The six-year operational dataset, together with the developed ML models, provides an empirical basis for interpreting the implications of the obtained findings for sustainable management of the analysed WWTP. Rather than representing a formal sustainability assessment, this section discusses the environmental, digital, and regulatory implications supported by the operational and modeling results. The long-term monitoring results

identify periods of increased environmental vulnerability. ML predictions can provide early warning of potential effluent-quality deterioration and support timely operator response. Together, these findings can support more reliable effluent-quality management and help reduce the risk of environmental and regulatory non-compliance. Figure 15 summarises how these findings relate to sustainable WWTP management. The organisation of the environmental and digital dimensions presented in Figure 15 is informed by previous studies addressing sustainable WWTP management, long-term operational monitoring, and data-driven approaches to wastewater treatment [5,6,25].

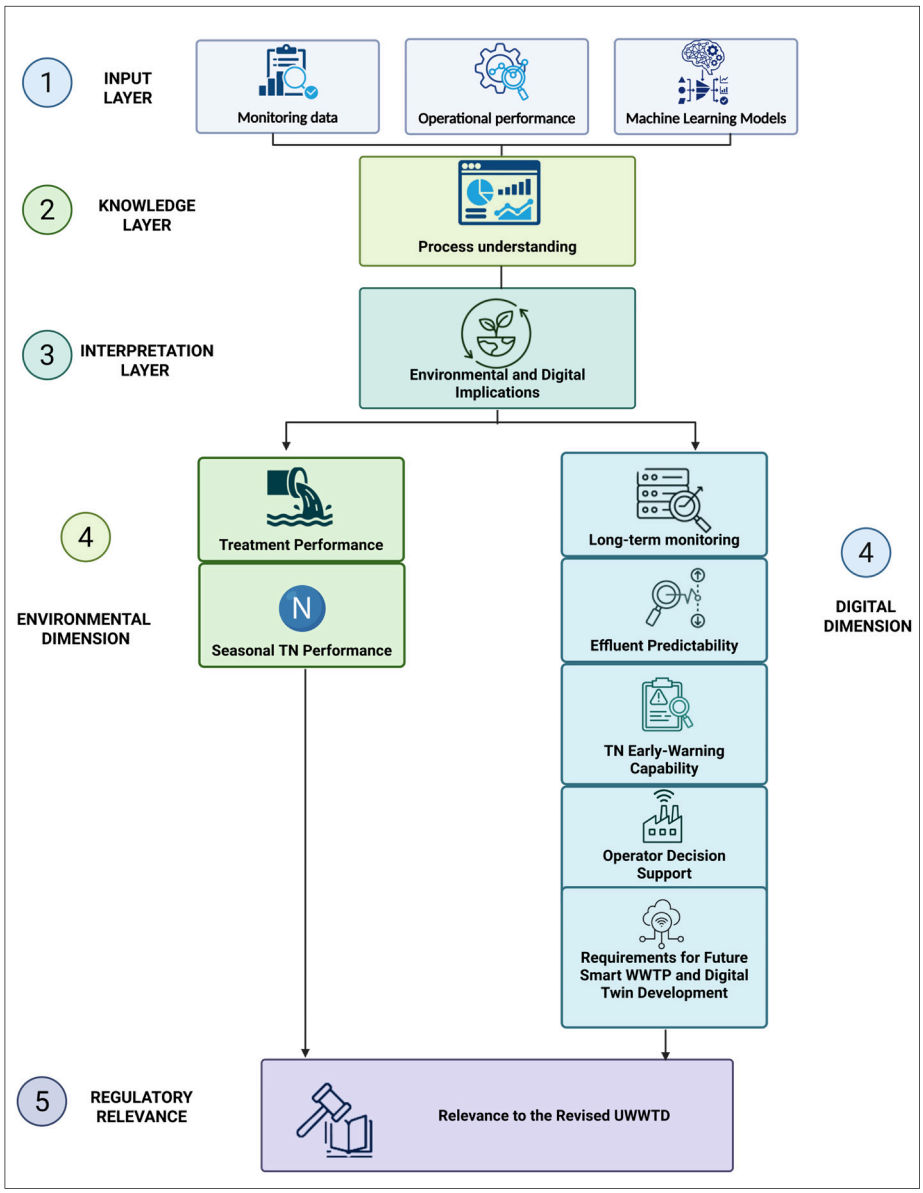


Figure 15. Conceptual representation of the environmental and digital dimensions relevant to the interpretation of the study findings for sustainable WWTP management. Figure created using BioRender.com.

3.9.1. Environmental Dimension

The long-term results showed a clear seasonal deterioration in biological treatment performance. TN removal decreased from 79.4% during the remainder of the year to 67.4% in summer (Section 3.5), while lower removal was also observed for BOD_5 and COD. In contrast, TSS and phosphorus removal remained relatively stable. Lower TN removal in summer was accompanied by more frequent effluent TN exceedances. These findings

show the importance of maintaining stable nitrogen removal during summer to ensure reliable effluent quality. For plant operators, summer is therefore a critical period for closer monitoring of nitrogen-removal performance and the risk of effluent non-compliance.

3.9.2. Digital Dimension

The six-year operational dataset provided the basis for evaluating the use of ML prediction in WWTP management. The practical value of the models became more evident at longer forecast horizons. This was particularly evident for TN and BOD_5, while COD predictions did not consistently outperform the baseline models.

For TN, the random forest achieved $R^2 = 0.516$ at a five-measurement forecast horizon, corresponding to approximately ten days. The R^2 values for the persistence and rolling-mean baselines were 0.388 and 0.320, respectively. Using 15 mg/L TN as the exceedance threshold, the model identified 49 of 69 exceedance days, with 12 false alarms (AUC = 0.896). At the selected decision threshold, this would correspond to approximately 61 alerts per year. About 80% of these alerts would represent actual exceedances.

The longer-horizon TN forecast could provide plant operators with advance information on effluent TN concentrations, while the exceedance classifier could support the identification of exceedance conditions. However, the SHAP analysis showed that, at the selected decision threshold of 0.5, the classifier primarily identified exceedance episodes that were already under way rather than reliably identifying their onset. Nitrogen-removal performance could then be monitored more closely. However, approximately 29% of exceedances would remain undetected. Routine monitoring therefore remains essential.

The model should be considered a decision-support tool rather than a basis for automated control. Further development toward Smart WWTP and Digital Twin applications would require higher-frequency monitoring and additional process measurements, particularly dissolved oxygen [34,35]. By providing predictive information on potential effluent-quality deterioration, ML models can support timely operational decisions and more reliable effluent management. This can contribute to more sustainable WWTP management.

3.9.3. Study Findings and the Revised UWWTD (2024)

The revised UWWTD provides a regulatory context for interpreting the study results. The Serbian regulatory threshold of 15 mg/L TN was exceeded on 60.8% of summer days and 38.5% of days during the remainder of the year (Section 3.5). These results show a higher risk of TN exceedance during summer, which is particularly relevant in view of the more stringent nitrogen-removal requirements introduced by the revised UWWTD (2024) [35].

For plant operators, these findings highlight the need for closer monitoring of nitrogen-removal performance during summer. Together with the ML-based early-warning results presented in Section 3.9.2, they can help identify periods with an increased risk of regulatory non-compliance and support timely process evaluation. The relationship between the study findings and the relevant priorities of the revised UWWTD is summarised in Figure S1 (Supplementary Material).

4. Conclusions

Six years of operational monitoring at a full-scale municipal wastewater treatment plant were used to characterise long-term performance and to benchmark machine learning for effluent forecasting against trivial baselines. Three findings emerge. The first is a systematic seasonal deterioration in biological treatment. Between June and August total nitrogen removal falls by 12.0 percentage points and mean effluent BOD_5 more than

doubles, in five of six monitored years, with no corresponding change in influent load or hydraulic flow.

The magnitude of the deficit orders as oxygen demand does, from nitrification through the oxidation of organic matter, while removal of suspended solids is entirely unaffected. That last observation excludes deterioration of sludge settling as an explanation and points to oxygen limitation, which is consistent with the roughly 20% fall in dissolved oxygen saturation across the observed temperature range. The inference is not confirmed directly, since the dataset contains no dissolved oxygen or aeration energy records, and obtaining them is the priority we recommend for future monitoring at this site. The finding is nonetheless actionable, identifying summer aeration capacity as the target for investigation. It is also relevant in a regulatory context, as exceedances of the Serbian 15 mg/L TN threshold occurred on 60.8% of summer days, while the revised UWWTD introduces more stringent nitrogen-removal requirements.

The second finding concerns how predictive performance should be measured. At a forecast horizon of one measurement, no parameter is predicted significantly better than both trivial baselines: total nitrogen exceeds a ten-measurement rolling mean but not persistence, while COD and BOD_5 exceed persistence but not the rolling mean. As the horizon lengthens the baselines decay rapidly, to R^2 of 0.001 for the rolling mean and 0.078 for persistence on nitrogen at ten steps, while model skill decays far more slowly. Two configurations exceed both baselines at conventional significance: total nitrogen at a horizon of five measurements using a random forest, and BOD_5 at ten measurements using either a random forest or an LSTM. COD is never predicted better than a rolling mean at any horizon. The one-step-ahead task that dominates the published literature is therefore both the least informative regime for assessing predictive value and the least useful operationally.

The third finding is methodological. Gap-filling a sparse operational record before evaluation inflates apparent performance from $R^2 = 0.34$ to $R^2 = 0.82$ over the same test period and can reverse the ranking of architectures. Approximately half of the daily series at this plant is not measured, and this is not unusual for operational monitoring records. We therefore recommend that studies report the measured fraction of their series, evaluate on measured observations, benchmark against both persistence and a rolling mean, and report performance at the forecast horizon at which the prediction would actually be used. The limitations of this study are its single-plant scope, daily rather than sub-daily sampling, and the absence of energy and chemical consumption data. The plant attenuates influent variability by factors between 2 and 15 depending on parameter, and that ordering matches the ordering of predictability, which suggests that predictive effort at conventional plants is better directed at weakly buffered parameters, such as nitrogen, than at the organic load indicators most commonly modeled.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/cleantechnol8050151/s1, Figure S1: Mapping the findings to the key priorities introduced by the revised Urban Wastewater Treatment Directive. Figure created using BioRender.com.

Author Contributions: Conceptualisation, A.V., N.J., I.M. and M.N.B.; methodology, N.J., S.I. and M.N.B.; software, N.J. and S.I.; validation, A.V., N.J., I.M., S.V., S.I. and M.N.B.; formal analysis, N.J. and S.I.; investigation, A.V., N.J., I.M., S.V., S.I. and M.N.B.; resources, S.V. and A.V.; data curation, S.V. and A.V.; writing—original draft preparation, A.V., N.J., I.M. and M.N.B.; writing—review and editing, A.V., N.J., I.M., S.V., S.I. and M.N.B.; visualisation, N.J., S.I. and M.N.B.; supervision, A.V. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The operational dataset and all analysis code supporting the reported results are publicly available at https://github.com/nikolaJovisic/upov-sombor (accessed on 3 September 2026). All results were generated with fixed random seeds and can be reproduced from that repository.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

WWTP	Wastewater Treatment Plant
COD	Chemical Oxygen Demand
ML	Machine Learning
DL	Deep Learning
LSTM	Long Short-Term Memory
PE	Population Equivalent
BOD_5	Biochemical Oxygen Demand (5-day)
TN	Total Nitrogen
TP	Total Phosphorus
TSS	Total Suspended Solids
Q	Wastewater Flow Rate
HRT	Hydraulic Retention Time
LR	Linear Regression
MLP	Multi-Layer Perceptron
ReLU	Rectified Linear Unit
Conv1D	One-Dimensional Convolutional Neural Network
SD	Standard Deviation
AUC	Area Under the Curve
RMSE	Root Mean Square Error
MAPE	Mean Absolute Percentage Error
CI	Confidence Interval
UWWTD	Urban Wastewater Treatment Directive
SHAP	Shapley Additive Explanations

References

1. Dutta, D.; Arya, S.; Kumar, S. Industrial wastewater treatment: Current trends, bottlenecks, and best practices. *Chemosphere* **2021**, *285*, 131245. [CrossRef]
2. Maryam, B.; Buyukgungor, H. Wastewater reclamation and reuse trends in Turkey: Opportunities and challenges. *J. Water Process Eng.* **2019**, *30*, 100501. [CrossRef]
3. Zhang, C.; Quan, B.; Tang, J.; Cheng, K.; Tang, Y.; Shen, W.; Su, P.; Zhang, C. China's wastewater treatment: Status quo and sustainability perspectives. *J. Water Process Eng.* **2023**, *53*, 103708. [CrossRef]
4. Tang, W.; Pei, Y.; Zheng, H.; Zhao, Y.; Shu, L.; Zhang, H. Twenty years of China's water pollution control: Experiences and challenges. *Chemosphere* **2022**, *295*, 133875. [CrossRef]
5. Newhart, K.B.; Holloway, R.W.; Hering, A.S.; Cath, T.Y. Data-driven performance analyses of wastewater treatment plants: A review. *Water Res.* **2019**, *157*, 498–513. [CrossRef]

6. Corominas, L.; Garrido-Baserba, M.; Villez, K.; Olsson, G.; Cortes, U.; Poch, M. Transforming data into knowledge for improved wastewater treatment operation: A critical review of techniques. *Environ. Model. Softw.* **2018**, *106*, 89–103. [CrossRef]
7. Verma, N.; Singh, G.; Ahsan, N. Water quality modeling-based assessment for the scope of wastewater treatment of the urban reach of River Yamuna at Delhi, India. *Environ. Monit. Assess.* **2024**, *196*, 155. [CrossRef]
8. Li, G.; Ji, J.; Ni, J.; Wang, S.; Guo, Y.; Hu, Y.; Liu, S.; Huang, S.F.; Li, Y.Y. Application of deep learning for predicting the treatment performance of real municipal wastewater based on one-year operation of two anaerobic membrane bioreactors. *Sci. Total Environ.* **2022**, *813*, 151920. [CrossRef]
9. Hassen, E.B.; Asmare, A.M. Predictive performance modeling of Habesha brewery wastewater treatment plant using artificial neural networks. *Chem. Int.* **2019**, *5*, 87–96. [CrossRef]
10. Bayat Varkeshi, M.; Mohammadi, K.; Najib, R. BOD and COD estimation in wastewater outflow via artificial neural network. In *Recent Advances in Environmental Science from the Euro-Mediterranean and Surrounding Regions*; Springer: Cham, Switzerland, 2018; pp. 875–876. [CrossRef]
11. Zhu, M.; Wang, J.; Yang, X.; Zhang, Y.; Zhang, L.; Ren, H.; Wu, B.; Ye, L. A review of the application of machine learning in water quality evaluation. *Eco-Environ. Health* **2022**, *1*, 107–116. [CrossRef]
12. Taoufik, N.; Boumya, W.; Achak, M.; Chennouk, H.; Dewil, R.; Barka, N. The state of art on the prediction of efficiency and modeling of the processes of pollutants removal based on machine learning. *Sci. Total Environ.* **2022**, *807*, 150554. [CrossRef]
13. Guo, H.; Jeong, K.; Lim, J.; Jo, J.; Kim, Y.M.; Park, J.P.; Kim, J.H.; Cho, K.H. Prediction of effluent concentration in a wastewater treatment plant using machine learning models. *J. Environ. Sci.* **2015**, *32*, 90–101. [CrossRef]
14. El-Rawy, M.; Abd-Ellah, M.K.; Fathi, H.; Ahmed, A.K.A. Forecasting effluent and performance of wastewater treatment plant using different machine learning techniques. *J. Water Process Eng.* **2021**, *44*, 102380. [CrossRef]
15. Mazraeh, A.; Bagherifar, M.; Shabanlou, S.; Ekhlasmand, R. A hybrid machine learning model for modeling nitrate concentration in water sources. *Water Air Soil Pollut.* **2023**, *234*, 721. [CrossRef]
16. Huang, Y.; Chen, Y. Prediction of total phosphorus based on distance correlation and machine learning methods—A case study of Dongjiang River, China. *Water Air Soil Pollut.* **2024**, *235*, 105. [CrossRef]
17. Zhong, H.; Yuan, Y.; Luo, L.; Ye, J.; Chen, M.; Zhong, C. Water quality prediction of MBR based on machine learning: A novel dataset contribution analysis method. *J. Water Process Eng.* **2022**, *50*, 103296. [CrossRef]
18. Nasir, N.; Kansal, A.; Alshaltone, O.; Barneih, F.; Sameer, M.; Shanableh, A.; Al-Shamma'a, A. Water quality classification using machine learning algorithms. *J. Water Process Eng.* **2022**, *48*, 102920. [CrossRef]
19. Wang, D.; Thunell, S.; Lindberg, U.; Jiang, L.; Trygg, J.; Tysklind, M.; Souihi, N. A machine learning framework to improve effluent quality control in wastewater treatment plants. *Sci. Total Environ.* **2021**, *784*, 147138. [CrossRef]
20. Wang, R.; Yu, Y.D.; Chen, Y.W.; Pan, Z.C.; Li, X.; Tan, Z.L.; Zhang, J.Q. Model construction and application for effluent prediction in wastewater treatment plant: Data processing method optimization and process parameters integration. *J. Environ. Manag.* **2022**, *302*, 114020. [CrossRef]
21. Miao, S.; Zhou, C.L.; Al Qahtani, S.A.; Alrashoud, M.; Ghoneim, A.; Lv, Z.H. Applying machine learning in intelligent sewage treatment: A case study of chemical plant in sustainable cities. *Sustain. Cities Soc.* **2021**, *72*, 103009. [CrossRef]
22. Zaghloul, M.S.; Achar, G. Application of machine learning techniques to model a full-scale wastewater treatment plant with biological nutrient removal. *J. Environ. Chem. Eng.* **2022**, *10*, 107430. [CrossRef]
23. Shanmugam, K.; Gadhamshetty, V.; Tysklind, M.; Bhattacharyya, D.; Upadhyayula, V.K.K. A Sustainable Performance Assessment Framework for Circular Management of Municipal Wastewater Treatment Plants. *J. Clean. Prod.* **2022**, *339*, 130657. [CrossRef]
24. Salles, R.; Mendes, J.; Baptista, A.J.; Moura, P. A Review of Indicators-Based Computational Methodologies to Improve the Sustainability of Wastewater Treatment Plants. *Comput. Chem. Eng.* **2025**, *202*, 109279. [CrossRef]
25. Cairone, S.; Hasan, S.W.; Choo, K.-H.; Lekkas, D.F.; Fortunato, L.; Zorpas, A.A.; Korshin, G.V.; Zarra, T.; Belgiorno, V.; Naddeo, V. Revolutionizing Wastewater Treatment toward Circular Economy and Carbon Neutrality Goals: Pioneering Sustainable and Efficient Solutions for Automation and Advanced Process Control with Smart Technologies. *J. Water Process Eng.* **2024**, *63*, 105486. [CrossRef]
26. Song, H.; Woo, T.; Kim, S.; Jeong, C.; Kim, M.; Heo, S.; Yoo, C. Predictive Monitoring of Wastewater Treatment Performance: Seasonal Microbial Activity and Data-Informed Water Quality Model. *J. Water Process Eng.* **2025**, *73*, 107718. [CrossRef]
27. Zhang, S.; Cao, J.; Gao, Y.; Sun, F.; Yang, Y. A Deep Learning Algorithm for Multi-Source Data Fusion to Predict Effluent Quality of Wastewater Treatment Plant. *Toxics* **2025**, *13*, 349. [CrossRef]
28. Cechinel, M.A.P.; Neves, J.; Fuck, J.V.R.; de Andrade, R.C.; Spogis, N.; Riella, H.G.; Padoin, N.; Soares, C. Enhancing Wastewater Treatment Efficiency through Machine Learning-Driven Effluent Quality Prediction: A Plant-Level Analysis. *J. Water Process Eng.* **2024**, *58*, 104758. [CrossRef]
29. Li, R.; Feng, K.; An, T.; Cheng, P.; Wei, L.; Zhao, Z.; Xu, X.; Zhu, L. Enhanced Insights into Effluent Prediction in Wastewater Treatment Plants: Comprehensive Deep Learning Model Explanation Based on SHAP. *ACS EST Water* **2024**, *4*, 1904–1915. [CrossRef]

30. Yu, Y.; Chen, Y.; Huang, S.; Wang, R.; Wu, Y.; Zhou, H.; Li, X.; Tan, Z. Enhancing the Effluent Prediction Accuracy with Insufficient Data Based on Transfer Learning and LSTM Algorithm in WWTPs. *J. Water Process Eng.* **2024**, *62*, 105267. [CrossRef]
31. Ramya, S.; Srinath, S.; Tuppad, P. Enhanced Wastewater Parameter Prediction Using Comprehensive Regression Framework: Leveraging Feature Engineering and Hyperparameter Optimization with Machine Learning. *Procedia Comput. Sci.* **2025**, *260*, 822–830. [CrossRef]
32. Republic of Serbia. Regulation on Emission Limit Values of Pollutants in Water and Deadlines for Their Achievement. Official Gazette of the Republic of Serbia, Nos. 67/2011, 48/2012 and 1/2016. Available online: https://www.paragraf.rs/propisi/uredba-granicnim-vrednostima-emisije-zagadjujucih-materija-u-vode.html (accessed on 25 August 2026).
33. European Parliament; Council of the European Union. Directive (EU) 2024/3019 of the European Parliament and of the Council of 27 November 2024 Concerning Urban Wastewater Treatment (Recast). Available online: https://eur-lex.europa.eu/eli/dir/2024/3019/oj (accessed on 3 September 2026).
34. Wang, A.-J.; Li, H.; He, Z.; Tao, Y.; Wang, H.; Yang, M.; Savic, D.; Daigger, G.T.; Ren, N. Digital Twins for Wastewater Treatment: A Technical Review. *Engineering* **2024**, *36*, 21–35. [CrossRef]
35. Li, Y.; Zhang, H.; Zhang, P.; Wang, Y.; Hao, Y.; Wang, X.; Guo, W. AI for Smart Wastewater Treatment Plants: A Review of Physics-Informed Water Quality Modeling, Optimization, and Advanced Control. *J. Environ. Manag.* **2026**, *401*, 128949. [CrossRef]

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